

Bayesian Statistics

The Bayesian Statistics course presents a practical introduction to the main principles and applications of modern statistical decision theory, with a special focus on Bayesian modeling, data analysis, inference, and optimal decision making. The course shows how three main sources of information – observed data, anticipated consequences of our decisions in the form of a loss function or risk, and knowledge before the experiment in the form of a prior distribution – are combined to produce optimal statistical solutions. Elaborated Bayesian methodology will be applied to real problems using statistical software, R and BUGS.

Time Wednesday, 5:30 – 8:00 pm

Room DMTI 109

Instructor [Michael Baron](#)

Office [DMTI 106-D](#)

Office hours Mon, Wed, Thu 4:00 – 5:00 pm

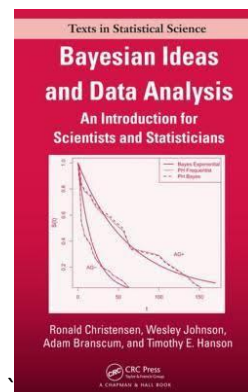
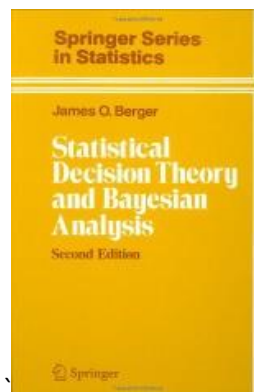
Phone 202-885-3130

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Course web site - <http://fs2.american.edu/baron/www/618/>

Textbooks No textbook is required; complete lecture notes will be given, based on parts of:

- **Statistical Decision Theory and Bayesian Analysis** by J. O. Berger, 2nd edition, 2010 or 1985, or 3rd edition, 1993 [classical text on the main concepts of decision theory, utility and risk, Bayesian, minimax, and sequential decisions]
- **Bayesian Ideas and Data Analysis. An Introduction for Scientists and Statisticians** by R. Christensen, W. Johnson, A. Branscum, and T. E. Hanson, 2011 [modern comprehensive guide to Bayesian fundamentals and modeling, with many practical examples, datasets, and detailed instructions in R and WinBUGS]



Course plan:

1. Introduction, motivation, and examples. Main principles and modern methods of statistical decision theory.
2. Bayesian philosophy: prior and posterior; comparison of frequentist and Bayesian approaches.
3. Bayesian inference: estimation, hypothesis testing, credible sets, and prediction.
4. Bayesian modeling and computation. Modern techniques, Markov chain Monte Carlo, Gibbs sampling, Metropolis-Hastings algorithm.
5. Bayesian computation in R. Introduction to Bayesian computing software BUGS (**B**ayesian inference **U**sing **G**ibbs **S**ampling).
6. Minimax decision rules. Making decisions against an intelligent opponent. Game theory.
7. Applications. Examples from economics, business, medicine, engineering, and other fields.

Software: We'll make use of R and WinBUGS. Advanced programming skills and computer knowledge are *not* required. Download R from <https://cran.r-project.org/> and/or WinBUGS following [instructions](#).

Course requirements: the final grade will be based on weekly homework, including occasional mini-projects, short but regular weekly quizzes, two midterm tests, and a final project.

Grading:

Weekly homework and mini-projects	10%	Homework will be assigned weekly and graded quickly. A steady effort to work out all the assigned problems is essential for learning Bayesian statistical methods and for the successful performance in this course. Homework solutions will be given on the course web site. A typical homework will include a few problems to do by hand, to see how things work, and later in the course, a few realistic problems to do using R or WinBUGS software.
Weekly quizzes	20%	10- to 15-minute quizzes in the end of every class. Each quiz covers the material of the preceding week and the latest homework. During a quiz, you may use one cheat sheet .
Two exams	25% each	Each midterm covers several chapters of the material. Taken in class; lecture notes are allowed . Time is limited – 1 hour 15 minutes.
Final project	20%	Each student will receive or choose a data set with data description, problem formulation, and instructions. Using sound Bayesian decision making strategies, the students will do the necessary modeling and data analysis and write a report summarizing results and answering specific questions of the project.

90 – 100 % = A

87 – 90 % = A-

83 – 87 % = B+

83 – 87 % = B

80 – 87 % = B-

77 – 80 = C+

73 – 77 % = C

77 – 80 % = C-

Prerequisites: Intermediate Statistics – STAT 302 or STAT 320 or STAT 514 or equivalent

Emergency Preparedness. In the event of an emergency, students should refer to the AU Web site (<http://www.american.edu/emergency>) and the AU information line at (202) 885-1100 for general university-wide information. In case of a prolonged closure of the University, I send updates to you by email and will post all announcements on the course web site.

Support Services. A wide range of services is available to support you in your efforts to meet the course requirements.

- *Mathematics & Statistics Tutoring Lab* (x3154, x3120, Don Myers Building, Room 103) provides tutoring in Mathematics and Statistics. Lab hours are Mo-Th 11 am – 8 pm, Fr 11 am – 3 pm, and Su 3 pm – 8 pm. <http://www.american.edu/cas/mathstat/tutoring.cfm>
- *Academic Support and Access Center* (x3360, MGC 243) offers study skills workshops, individual instruction, tutor referrals, Supplemental Instruction, writing support, and technical and practical support and assistance with accommodations for students with physical, medical, or psychological disabilities. Writing support is also available in the Writing Center, Battelle-Tompkins 228.
- *CTRL Connect* – [software support with R](mailto:ctrl@american.edu) (ctrl@american.edu, x2117)
- *Counseling Center* (x3500, MGC 214) offers counseling and consultations regarding personal concerns, self-help information, and connections to off-campus mental health resources.

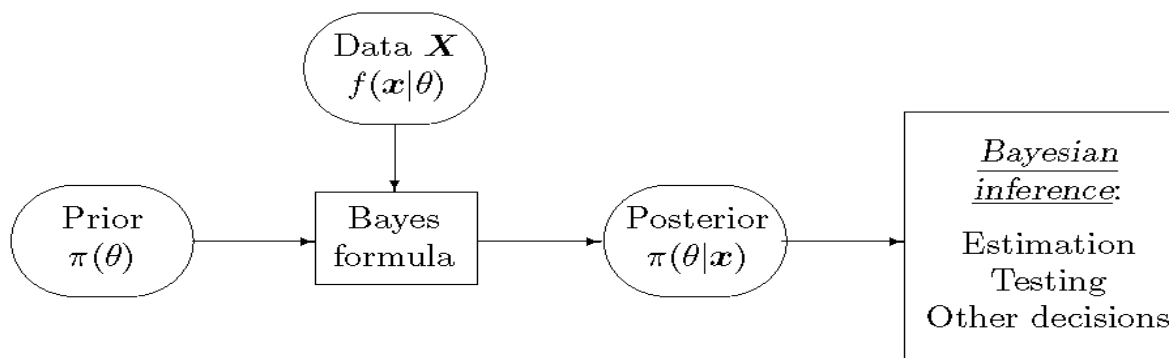
What are Bayesian Statistics and Decision Theory all about?

Statisticians deal with **data**. Most of the statistical decisions are based on carefully collected and processed samples of data. However, in our dynamic world, other sources of information cannot be neglected when we try to make optimal and accurate decisions. What other sources can we use, besides objective data?

The classical *decision theory* uses at least *two* kinds of knowledge: data and anticipated consequences of our decisions, in the form of a **loss function** or **risk**. For example, if we underestimate and therefore, underprice a product, we lose the dollar difference. On the other hand, if we overprice it, we may lose customers, which is a completely different outcome. In general, risk is the expected loss, and our decision is optimal – in the frequentist, Bayesian, minimax, or other sense – if it minimizes some form of a risk.

In addition, *Bayesian decision theory* uses the *third* kind of information – what we know before the experiment, i.e., the **prior distribution**. Bayesians treat unknown parameters as random variables and express the prior knowledge about them in a form of a distribution. As part of “Bayesian philosophy”, we’ll discuss many reasons for doing so. Moreover, we’ll see a number of situations that can only be explained from a Bayesian point of view.

What happens next? Using the elementary Bayes Rule, the prior distribution gets combined with the data to yield the **posterior distribution** – our best knowledge about parameters after the experiment, after we saw the data. All Bayesian decisions are based on the posterior.



When new data arrive, they are incorporated into the same scheme. The old posterior distribution plays the role of a prior, and through the Bayes formula, it gets converted into the updated posterior, just like our auto insurance premium gets modified every year.

Bayesian inference is not the only way of making statistical decisions. The Bayes risk **averages** our possible losses over the prior distribution. But in a competitive world, we may worry about the **worst** possible risk rather than the average. When we act against an *intelligent opponent* such as a market competitor, this opponent will try to create the most beneficial situation for him which will likely be the least beneficial for us. Then we should choose **minimax** decisions, which minimize the maximum possible risk. These are conservative strategies which ensure that even in the worst case, we do not lose too much. **Game Theory** studies such strategies and how to calculate them.

Further reading: This NY Times article explains the Bayesian approach and how it saved one fisherman’s life: <http://www.nytimes.com/2014/09/30/science/the-odds-continually-updated.html>